

Algebra 1
Unit 11
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ growth factor and b represents ☒ initial amount

the ☒ growth factor.
☐ initial amount.

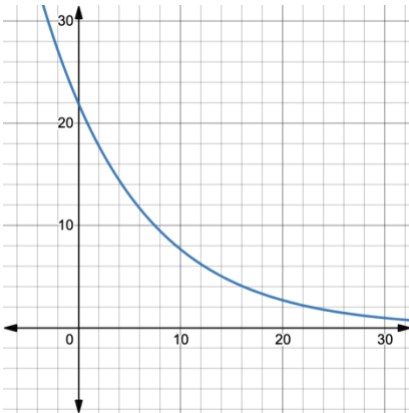
2. When b is between 0 and 1, the equation models ☐ exponential growth.
☒ exponential decay.

3. When b is greater than 1, the equation models ☒ exponential growth.
☐ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4. $y = 142(1.05)^x$ represents exponential growth because...

$b > 1$ and percent rate of change is greater than 100%.

5.  represents exponential decay because... y -values decrease while x -values increase.

6. represents exponential growth because...

x	y
-1	$\frac{1}{5}$
0	$\frac{1}{2}$
1	$\frac{5}{4}$
2	$\frac{25}{8}$

increasing at a rate 2.5 times the previous value
(a factor of 2.5)

$\times 2.5$

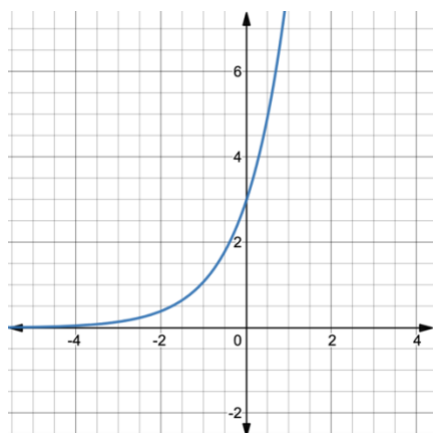
$\times 2.5$

$\times 2.5$

7. $y = 3(1 - 0.14)^x$ represents exponential decay because...

$3(.86)^x$ the value of b is < 1 .

8. represents exponential growth because...



The graph increases at a rapid rate. As x increases, y increases at a rapid rate.

9. Write an equation in $y = a(b)^x$ form that represents exponential growth.

$$y = 2(1.25)^x$$

$$b > 1$$

10. Write an equation in $y = a(b)^x$ form that represents exponential decay.

$$y = 2(.94)^x$$

$$0 < b < 1$$

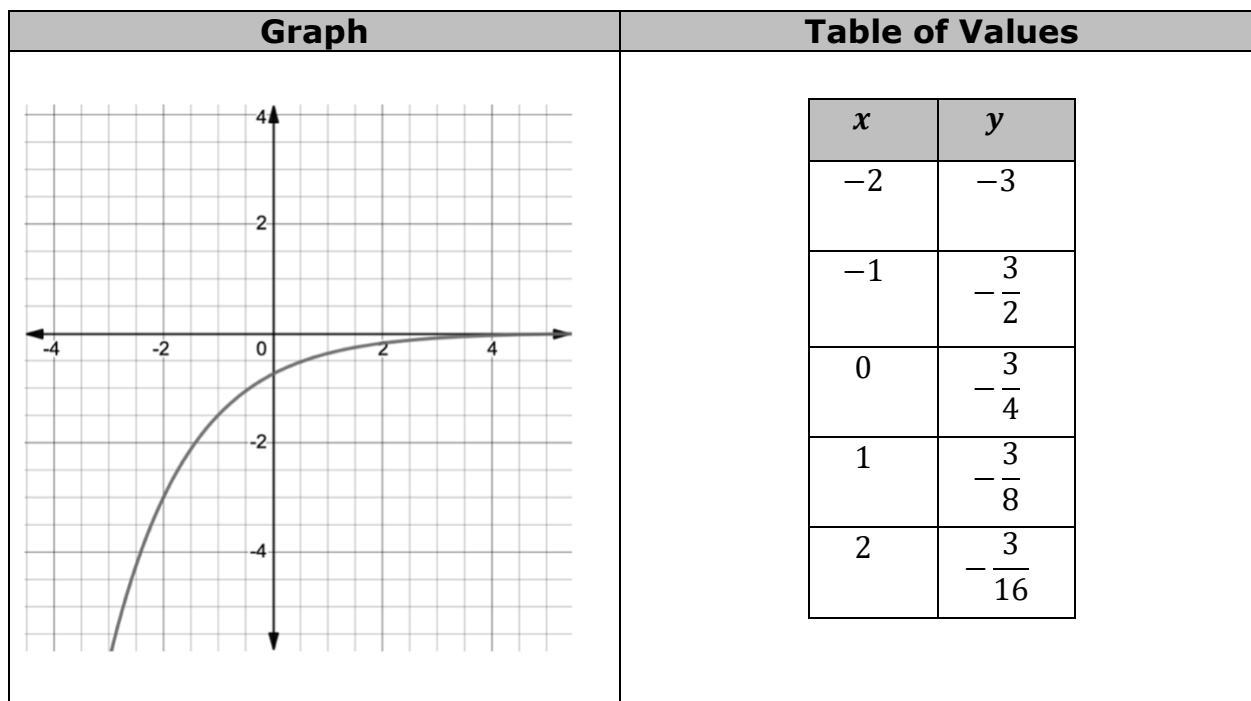
Algebra 1
Unit 11
Lesson 2 Practice

Name Answer Key
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1. The graph of $y = -1(1.7)^x$ is ☐ increasing and is ☐ positive.
☒ decreasing ☒ negative.

2. The graph of $y = \left(\frac{7}{3}\right)^x$ is ☒ increasing and is ☒ positive.
☐ decreasing ☐ negative.

Use the graph and table of $y = -\frac{3}{4}(1 - 0.5)^x$ to determine key features.



Key Features			
3. Domain	All real numbers	4. Range	$y < 0$
5. y- intercept	$(0, -\frac{3}{4})$	6. Constant Percent Rate of Change	$.5(100) = 50\%$
7. Increasing or Decreasing	Increasing $-\infty < x < \infty$	8. Positive	Never positive
		Negative	$-\infty < x < \infty$
9. Left End Behavior	$x \rightarrow -\infty, y \rightarrow -\infty$	10. Right End Behavior	$x \rightarrow \infty, y \rightarrow 0$

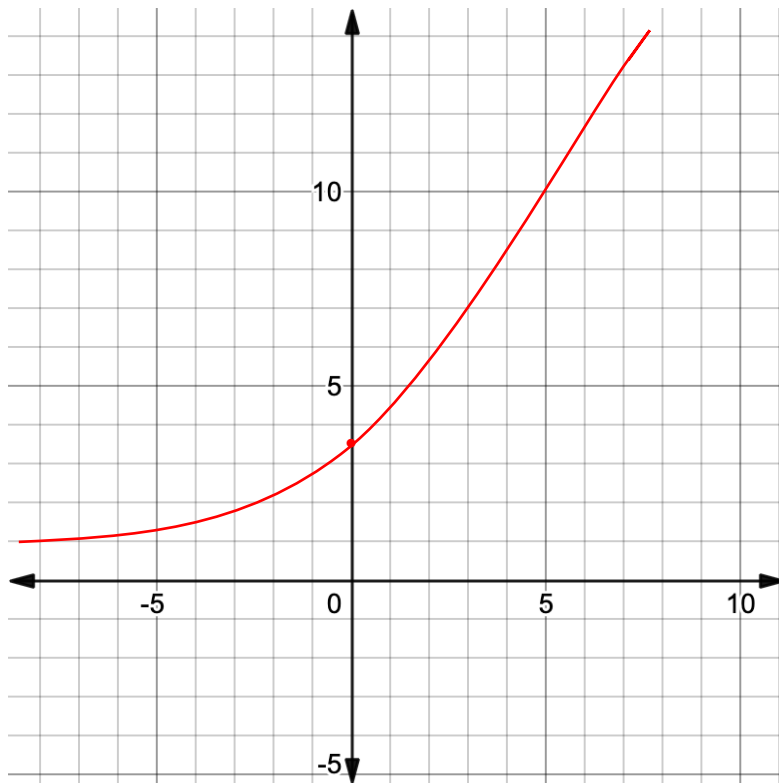
Algebra 1
Unit 11
Lesson 3 Practice

Name **Answer Key** _____
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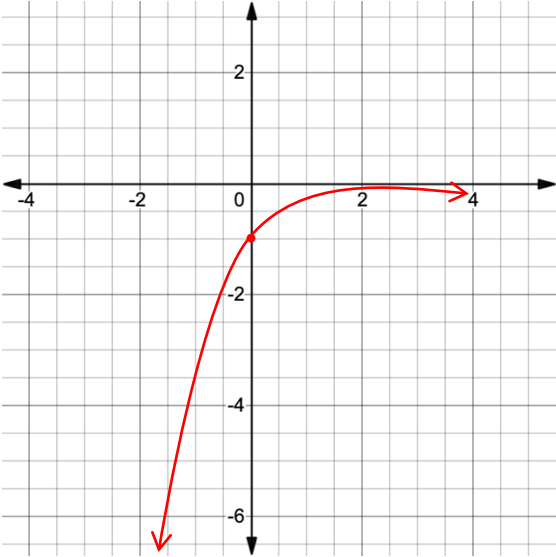
For questions 1-2, consider the function $f(x) = \frac{7}{2}(1.23)^x$.

1. Since a is ☒ positive
☐ negative and b is ☐ between 0 and 1,
☒ greater than 1, the function
- is ☒ increasing.
☐ decreasing. The left end behavior of the graph is ☒ $y \rightarrow 0$
☐ $y \rightarrow \infty$ and the
- right end behavior is ☐ $y \rightarrow 0$.
☒ $y \rightarrow \infty$. The y -intercept of the function is ☐ 0.23.
☐ 1.23.
☒ 3.5.

2. Sketch the graph of $f(x) = \frac{7}{2}(1.23)^x$.



Graph the function of $y = -(1 - 0.75)^x$, complete the table of values, and then determine all key features.

3. Graph	4. Table of Values												
	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-16</td></tr><tr><td>-1</td><td>-4</td></tr><tr><td>0</td><td>-1</td></tr><tr><td>1</td><td>$-\frac{1}{4}$</td></tr><tr><td>2</td><td>$-\frac{1}{16}$</td></tr></table>	x	y	-2	-16	-1	-4	0	-1	1	$-\frac{1}{4}$	2	$-\frac{1}{16}$
x	y												
-2	-16												
-1	-4												
0	-1												
1	$-\frac{1}{4}$												
2	$-\frac{1}{16}$												

Key Features	
5. Growth or Decay	decay since b is .25 which is less than 1
6. Domain and Range	Domain: $\mathbb{R}$ Range: $y < 0$
7. Increasing/Decreasing	increasing $-\infty < x < \infty$
8. Constant Percent Rate of Change	.25 = 25%
9. Left and Right End Behavior	left: $x \rightarrow -\infty$ $y \rightarrow -\infty$ right: $x \rightarrow \infty$ $y \rightarrow 0$
10. Positive or Negative	always negative $-\infty < x < \infty$

Algebra 1
Unit 11
Lesson 4 Practice

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Use the scenario below for questions 1-5.

Janice started a bank account for her daughter with \$625, which earns 4.75% interest per year. The amount in her daughters account after x years can be modeled by the function $V(x) = \$625(1.0475)^x$.

1. Does the equation $V(x) = \$625(1.0475)^x$ represent exponential growth or decay?

exponential growth since $b > 1$

2. What is the y -intercept and what does it represent in this situation?

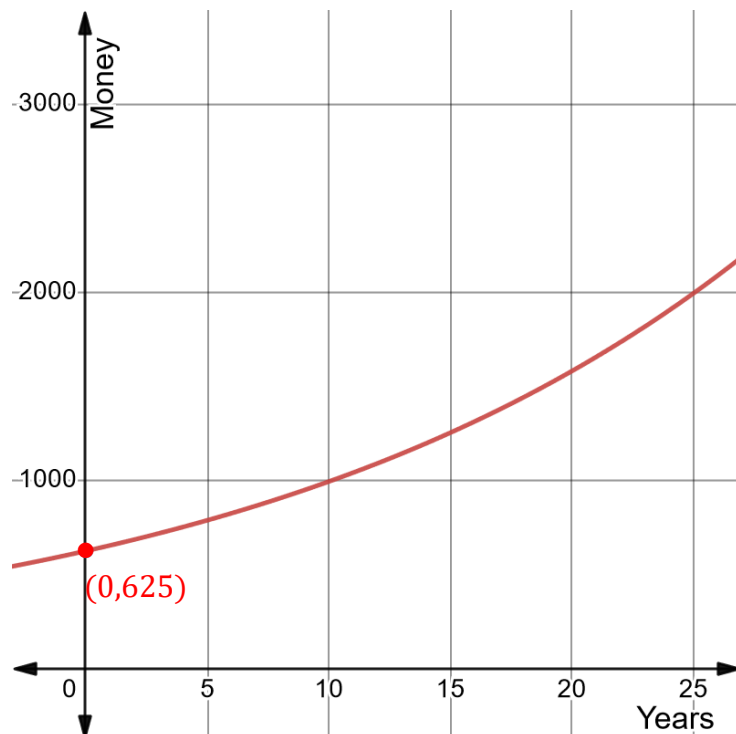
y -intercept is 625. This represents the initial amount deposited.

3. Explain why the value of b is not 4.75, the interest rate.

The interest rate is a percent. $4.75\% = \frac{4.75}{100} = .0475$.

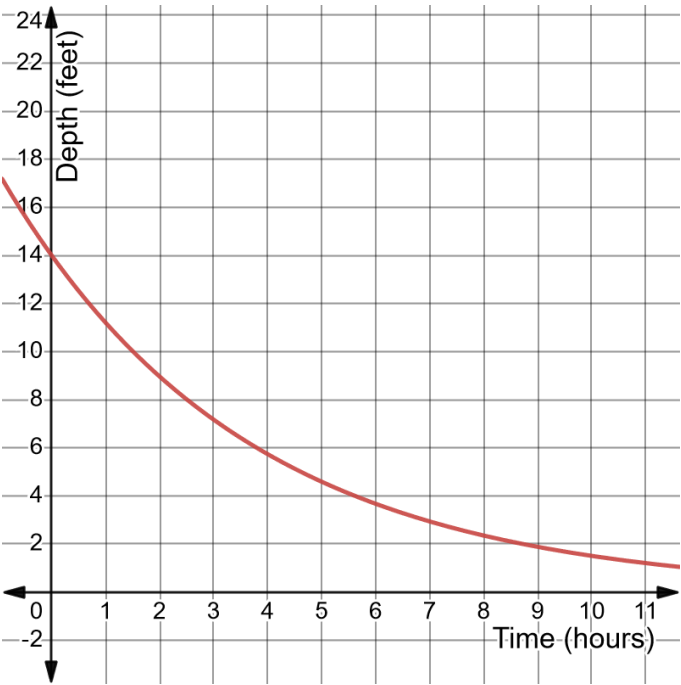
This is then added to the 100% which was deposited originally.

4. Graph the equation $V(x) = \$625(1.0475)^x$.



The equation $y = 14\left(\frac{4}{5}\right)^x$ represents the depth of the water as a pool drains in feet in x hours. Graph the function below, then determine the key features and interpret how each relates to the context.

5.

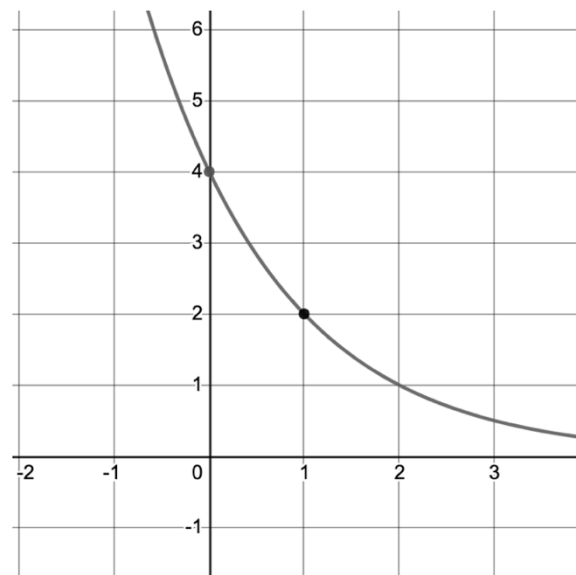


Key Features		Interpretations
6. Domain	$x \geq 0$	The domain is the time the pool drains for
7. Range	$0 \leq y \leq 14$	At time 0 hours, the pool is 14 feet deep until no water is left
8. y -intercept	$(0,14)$	The initial depth of the water in the pool
9. Constant Percent Rate of Change	$y = 14(1 - .2)^x$ $14(.8)^x$	Rate at which the pool is draining every hour
10. End Behavior	$x \rightarrow \infty, y \rightarrow 0$	As time continues, the pool drains until no water is left.

Algebra 1
Unit 11
Lesson 5 Practice

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The graph of an exponential function is shown.



1. Find the value of a by substituting the point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{4} = a(b)^{\boxed{0}} \quad a = 4$$

2. Use the value of a found in question 1 and the point $(1, 2)$ to determine the value of b .

$$y = 4(b)^x$$

$$\frac{2}{4} = \frac{4(b)^1}{4} \quad b = \frac{1}{2}$$

3. Write the function for the graph in both forms.

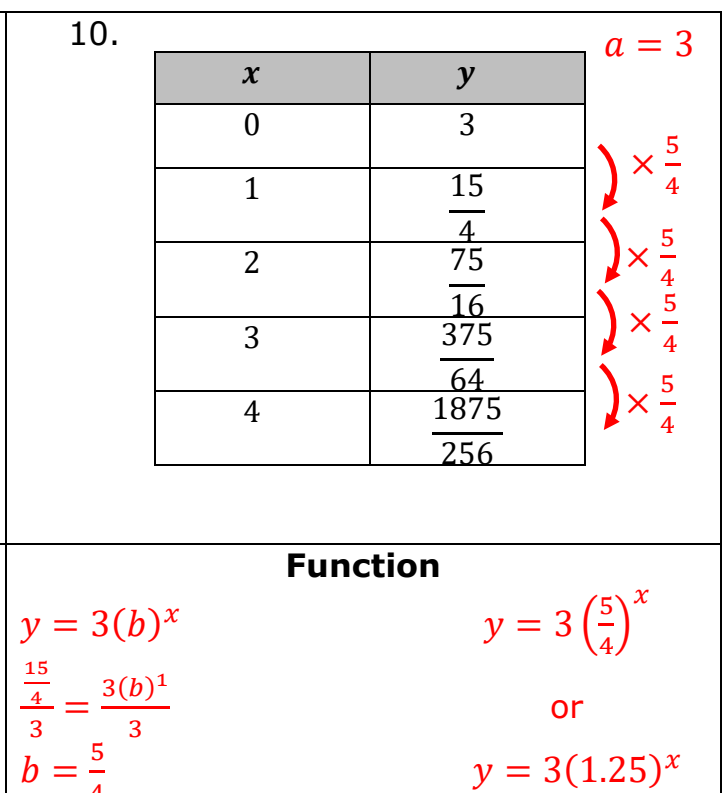
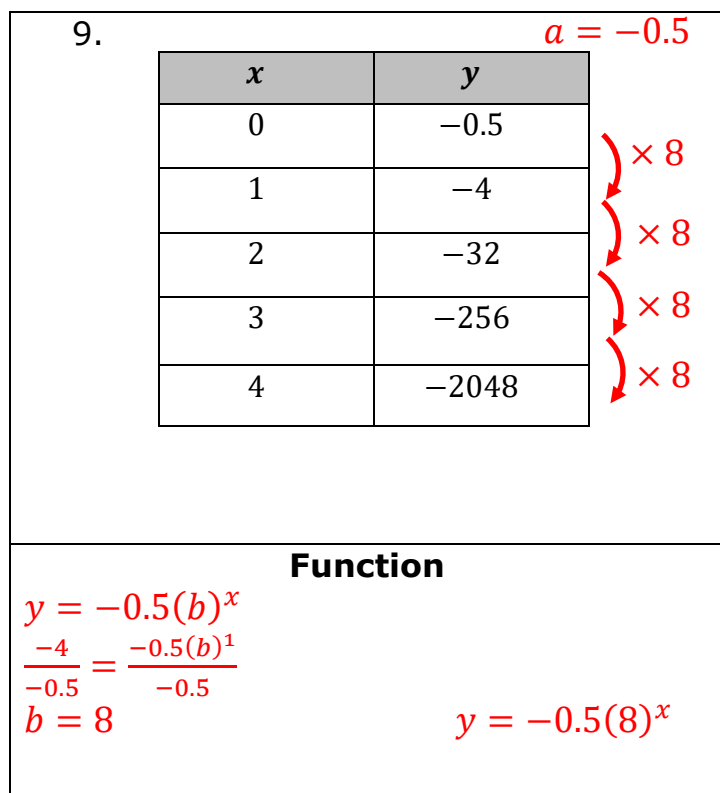
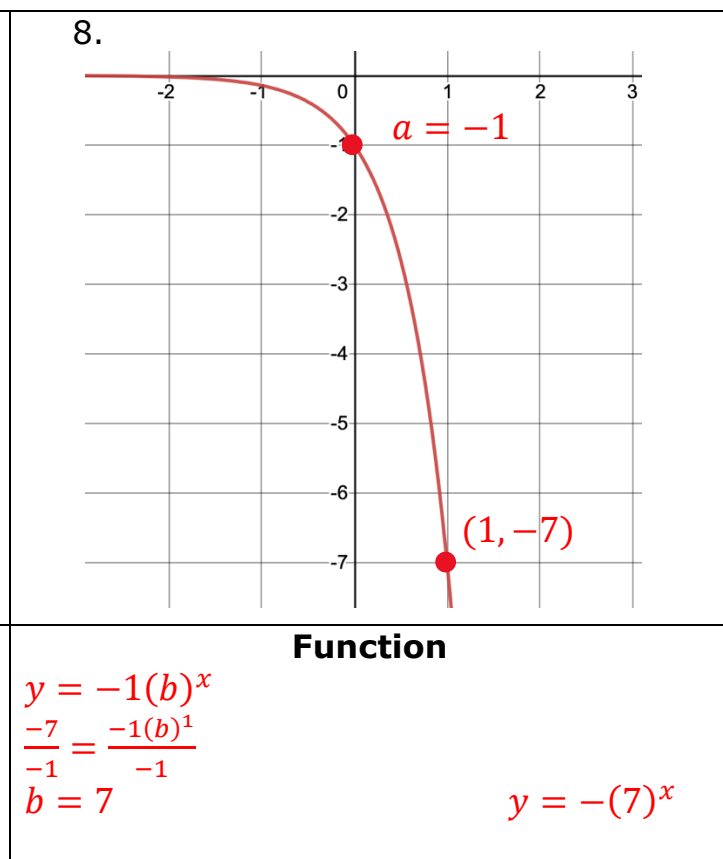
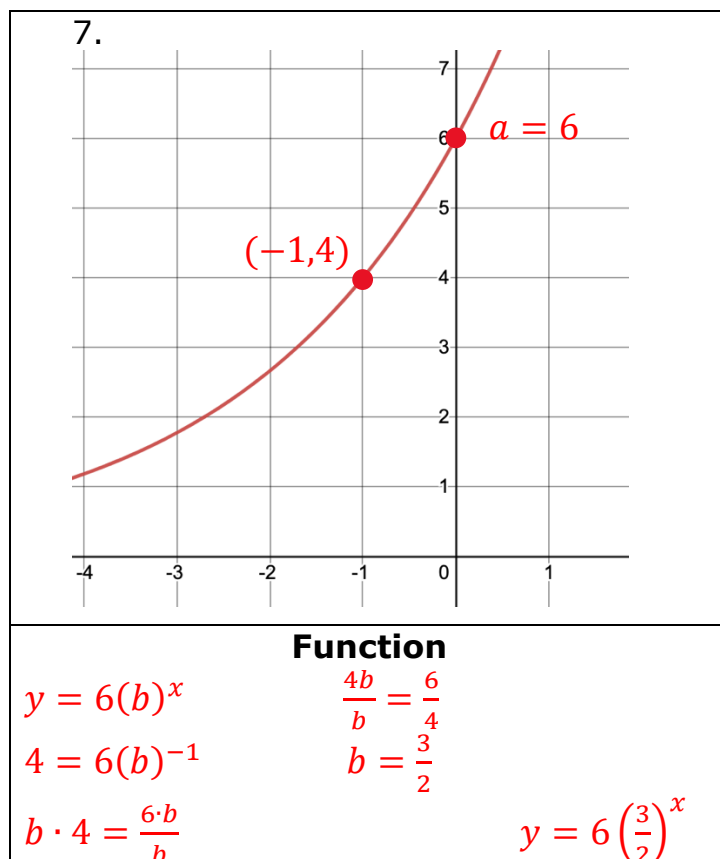
$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$
$f(x) = 4\left(\frac{1}{2}\right)^x$	$f(x) = 4(1 - 0.5)^x$

The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	0.4	1.2	3.6	10.8	32.4

4. What is the initial value, a ?
 $a = 0.4$
5. How does the value of $f(x)$ change as the x -value increases by 1?
As the x -value increases by 1, the y -value or $f(x)$ value is multiplied by a factor of 3.
6. Write the function in the form $f(x) = a(b)^x$
 $f(x) = 0.4(3)^x$

For questions 7-10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .



Algebra 1
Unit 11
Lesson 6 Practice

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Use the situation below to answer questions 1-3.

A new MacBook computer costs \$950, but depreciates 10.8% each year as new technology comes out.

1. Is this situation exponential growth or decay?
Exponential decay

2. Two students, Quinn and Jyquis were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Quinn	Jyquis
$V(x) = 950(1.108)^x$	$V(x) = 1275(0.892)^x$

I agree with neither of them. Quinn has exponential growth in her equation, and Jyquis has the incorrect initial amount.

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$
$V(x) = 950(0.892)^x$	$V(x) = 950(1 - 0.108)^x$

Use the situation below to answer questions 4-6.

A new school opened with 134 students in 2021 and plans to increase by 26% per year until they reach full capacity.

4. Is this situation exponential growth or decay?
Exponential growth

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.

$$P(x) = 134(1.26)^x \text{ or } P(x) = 134(1 + 0.26)^x$$

6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$
$P(x) = 134(1.26)^x$	$P(x) = 134(1 + 0.26)^x$

7. A three-bedroom house in the suburbs of Miami sold for \$420,625. If the housing prices are expected to increase 3.75% annually, write a function that models the cost of a house, C , x years after it was sold.

$$C(x) = 420,625(1 + 0.0375)^x \text{ or } C(x) = 420,625(1.0375)^x$$

8. A small local company reported a profit of \$89,000 last year but is struggling due to the pandemic. They are reporting a decrease of 12% each year. Write a function that represents the profit, P , after t years.

$$P(t) = 89,000(1 - 0.12)^t \text{ or } P(t) = 89,000(0.88)^t$$

9. As of 2020, the world population is 9.23 billion people and is expected to grow at a rate of 0.62% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.

$$P(y) = 9.23(1 + 0.0062)^y \text{ or } P(y) = 9.23(1.0062)^y$$

10. A culture of 15 bacteria grows at a rate of 250% each hour. Tyler wrote the equation $B(t) = 15(2.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error that Tyler made.

Tyler forgot to add the 1 or 100% to the growth rate.

$$B(t) = 15(1 + 2.5)^t \text{ or } B(t) = 15(3.5)^t$$